C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
`http://c-immsim.iac.rm.cnr.it (alias to http://kraken.iac.rm.cnr.it/C-IMMSIM)`

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
tt doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
`https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262i`

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
`https://jitc.bmj.com/content/9/5/e002694`

Original C-IMMSIM model: `www.iac.cnr.it/∼filippo/c-immsim`

GETTING HELP:
Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)
Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

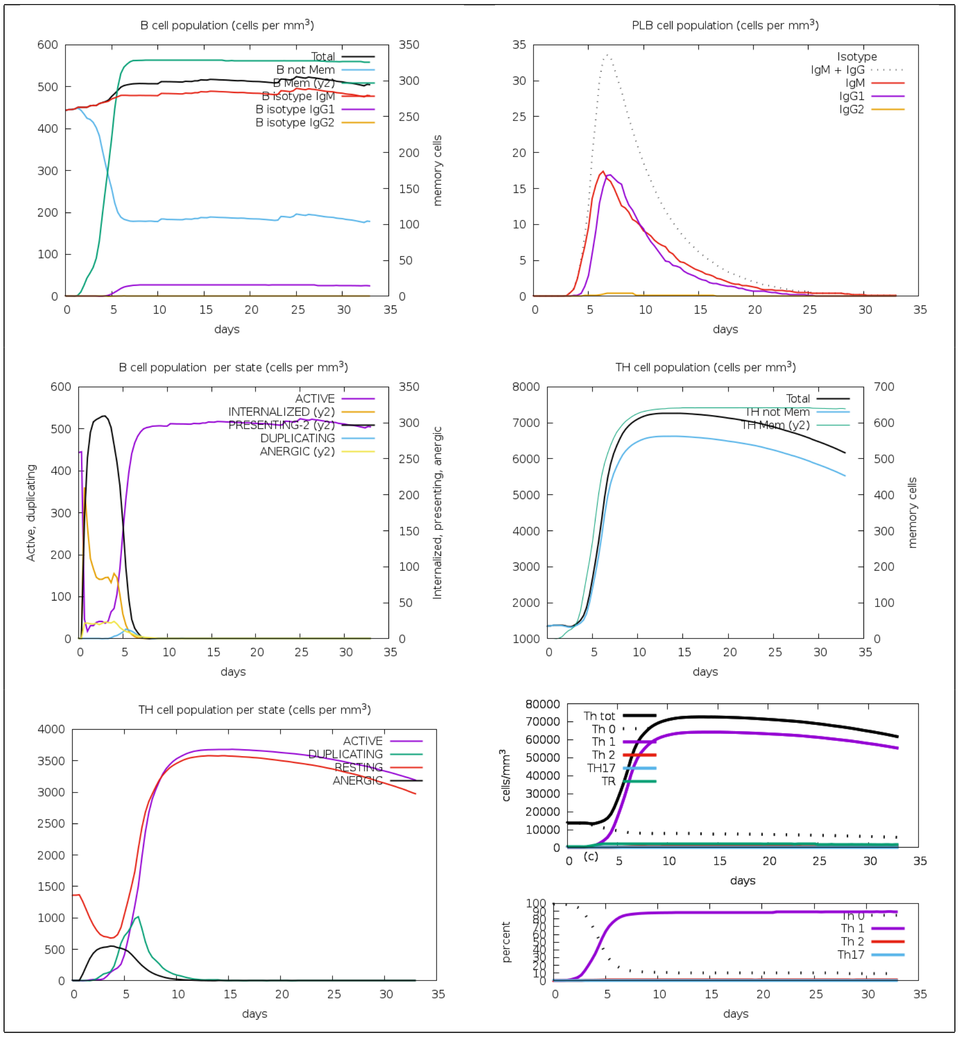

Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

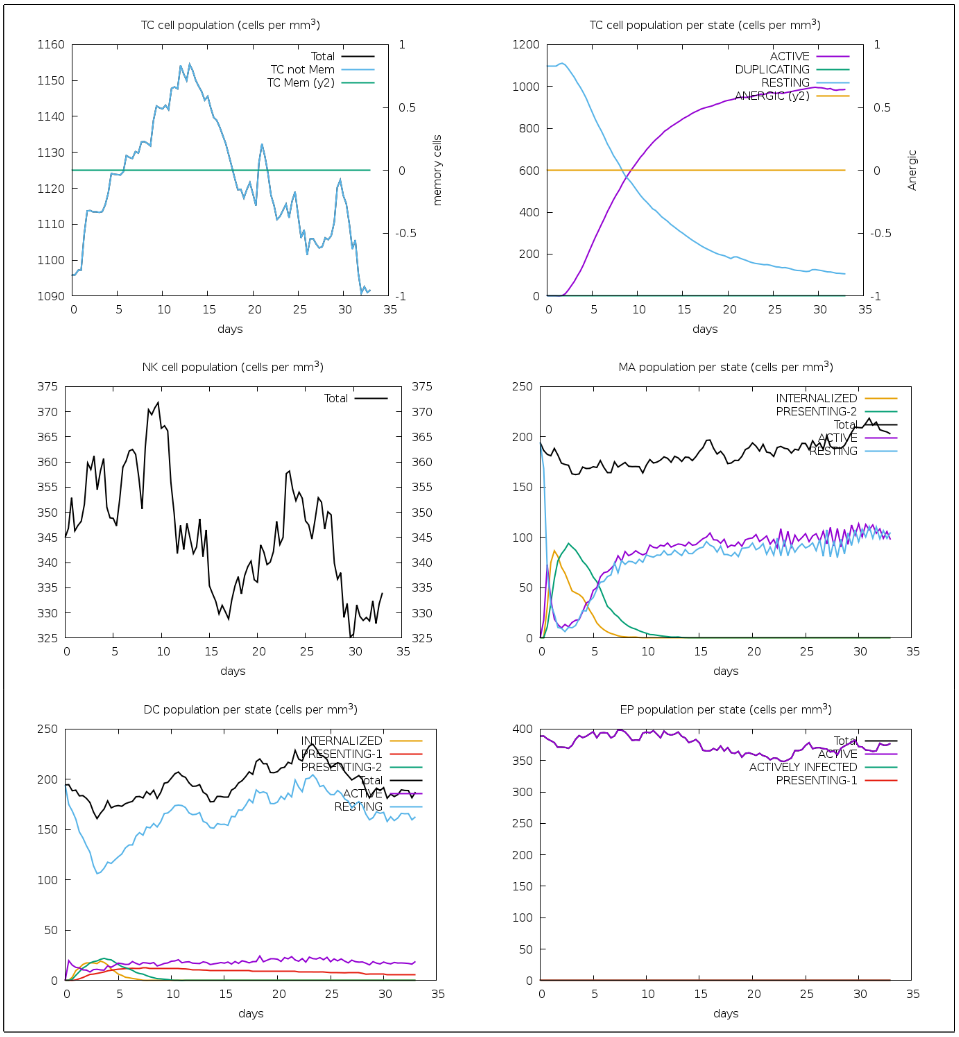

Figure 2: Legend: symbols as figure above.

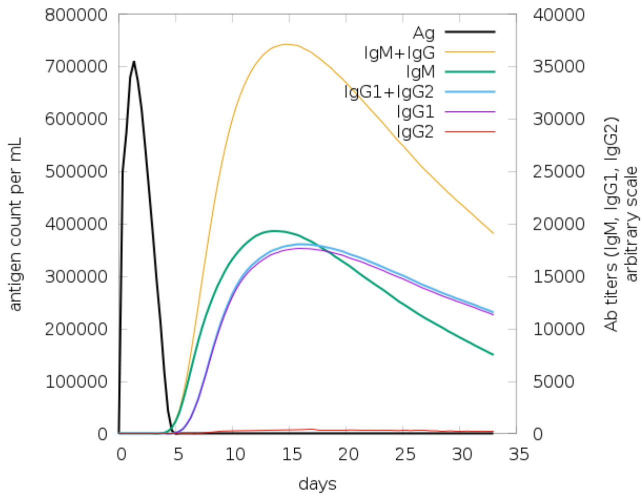

Figure 3: The virus, the immunoglobulins and the immunocomplexes.

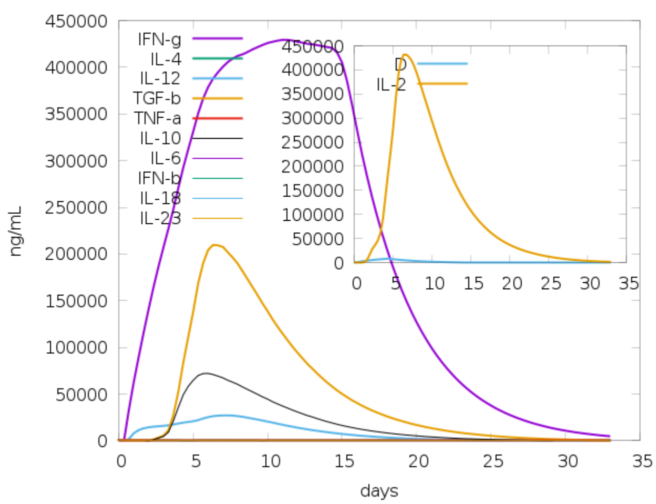

Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

```
Parker_B_scale_epitope_find
Use Parker’s propensity scale, takes an antigen block as input,
and creates a list of residues that are possible epitopes.


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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020212_5_8yRPooZoQxXK.FSA_1_001
----------------------------------


AKFVAAWTLKAAAGGSANQGQQLADIGVPQPAPVAAPARRTRKPQQPESLEECDSELEIKRYKNRVASEAAAKGPGTGPG
NGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGA
GAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

00000000000010111110010000000000000000011111110110111110000000000100001001111110
00111111111111111111111111111111111111111111111111111110111100111000000011111111
11111111111111111111111111111111111111111111111111111111110000000000000000000000
0000000000000000000000010000000000000000000000000000000000000011111000000000
Epitopes of protein 0 ------------
1]       pos=14 len=5        GSANQ
2]       pos=39 len=7        RTRKPQQ
3]       pos=50 len=5        EECDS
4]       pos=73 len=6        GPGTGP
5]       pos=82 len=53        LGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRH
6]       pos=136 len=4        DGVR
7]       pos=152 len=66        GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAG
8]       pos=302 len=5        TQGPE
```

```
DoPeptideList_I:
        Given the antigen injected creates the list of peptides for all the
        NumAgProts proteins and for all i.e., 4 MHCI molecules

        Read class I peptide list from file? NO

========================================================================

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000


----------------------------------
Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020212_5_8yRPooZoQxXK.FSA_1_001
----------------------------------

AKFVAAWTLKAAAGGSANQGQQLADIGVPQPAPVAAPARRTRKPQQPESLEECDSELEIKRYKNRVASEAAAKGPGTGPG
NGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGA
GAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

Epitopes of protein 0 ------------
0]       pos=  -1 score=1.000000 unnormalised=104.8410000000        non-binding event

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```

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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020212_5_8yRPooZoQxXK.FSA_1_001
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AKFVAAWTLKAAAGGSANQGQQLADIGVPQPAPVAAPARRTRKPQQPESLEECDSELEIKRYKNRVASEAAAKGPGTGPG
NGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGA
GAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

Epitopes of protein 0 ------------
0]        pos=  -1 score=1.000000 unnormalised=104.8410000000        non-binding event

=========================================================================

Allele: B0702
Pseudo sequence: KAAREEQQIKAQTRE
Threshold: 8.702800
Max score: 28.406000


----------------------------------
Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020212_5_8yRPooZoQxXK.FSA_1_001
----------------------------------

AKFVAAWTLKAAAGGSANQGQQLADIGVPQPAPVAAPARRTRKPQQPESLEECDSELEIKRYKNRVASEAAAKGPGTGPG
NGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGA
GAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

Epitopes of protein 0 ------------
0]        pos=  27 score=0.033098 unnormalised=3.8422000000        VPQPAPVAA
1]        pos=  29 score=0.031022 unnormalised=3.6012000000        QPAPVAAPA
2]        pos= 121 score=0.002595 unnormalised=0.3012000000        RPGAPGGSG
3]        pos= 140 score=0.016645 unnormalised=1.9322000000        RPQKRPSCI
4]        pos= 264 score=0.013500 unnormalised=1.5672000000        GPGLPQGQL
5]        pos=  -1 score=0.903140 unnormalised=104.8410000000        non-binding event

=========================================================================

Allele: B0702
Pseudo sequence: KAAREEQQIKAQTRE
Threshold: 8.702800
Max score: 28.406000


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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020212_5_8yRPooZoQxXK.FSA_1_001
----------------------------------

AKFVAAWTLKAAAGGSANQGQQLADIGVPQPAPVAAPARRTRKPQQPESLEECDSELEIKRYKNRVASEAAAKGPGTGPG
NGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGA
GAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

Epitopes of protein 0 ------------
0]        pos=  27 score=0.033098 unnormalised=3.8422000000        VPQPAPVAA
1]        pos=  29 score=0.031022 unnormalised=3.6012000000        QPAPVAAPA
2]        pos= 121 score=0.002595 unnormalised=0.3012000000        RPGAPGGSG
3]        pos= 140 score=0.016645 unnormalised=1.9322000000        RPQKRPSCI
4]        pos= 264 score=0.013500 unnormalised=1.5672000000        GPGLPQGQL
```

```
5]       pos=  -1 score=0.903140 unnormalised=104.8410000000        non-binding event
```

```
DoPeptideList_II:
        Given the antigen injected creates the list of peptides for all the
        NumAgProts proteins and for all i.e., 2 MHCII molecules

        Read class II peptide list from file? NO

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Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000


----------------------------------
Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020212_5_8yRPooZoQxXK.FSA_1_001
----------------------------------

AKFVAAWTLKAAAGGSANQGQQLADIGVPQPAPVAAPARRTRKPQQPESLEECDSELEIKRYKNRVASEAAAKGPGTGPG
NGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGA
GAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

Epitopes of protein 0 ------------
0]       pos=   2 score=0.038281 unnormalised=3.3845600000        FVAAWTLKA
1]       pos=   6 score=0.005220 unnormalised=0.4615600000        WTLKAAAGG
2]       pos=   8 score=0.065517 unnormalised=5.7925600000        LKAAAGGSA
3]       pos=  61 score=0.018793 unnormalised=1.6615600000        YKNRVASEA
4]       pos=  65 score=0.009609 unnormalised=0.8495600000        VASEAAAKG
5]       pos= 229 score=0.018929 unnormalised=1.6735600000        VVLLGPGPG
6]       pos= 230 score=0.009632 unnormalised=0.8515600000        VLLGPGPGT
7]       pos= 246 score=0.033135 unnormalised=2.9295600000        FGPGPGACM
8]       pos= 259 score=0.010084 unnormalised=0.8915600000        LEKGPGPGL
9]       pos= 300 score=0.042409 unnormalised=3.7495600000        VTTQGPEAA
10]       pos=  -1 score=0.748392 unnormalised=66.1680000000        non-binding event

=========================================================================

Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000


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Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-020212_5_8yRPooZoQxXK.FSA_1_001
----------------------------------

AKFVAAWTLKAAAGGSANQGQQLADIGVPQPAPVAAPARRTRKPQQPESLEECDSELEIKRYKNRVASEAAAKGPGTGPG
NGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGA
GAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEAAAKRVLSVVVLLGPGPGTM
FEVSVAFGPGPGACMLSAPLEKGPGPGLPQGQLTAYGPGPGFLDKGTYTLGPGPGELRLNVTTQGPEAAAKPPMVE

Epitopes of protein 0 ------------
0]       pos=   2 score=0.038281 unnormalised=3.3845600000        FVAAWTLKA
1]       pos=   6 score=0.005220 unnormalised=0.4615600000        WTLKAAAGG
2]       pos=   8 score=0.065517 unnormalised=5.7925600000        LKAAAGGSA
3]       pos=  61 score=0.018793 unnormalised=1.6615600000        YKNRVASEA
```

```
4]        pos=  65 score=0.009609 unnormalised=0.8495600000        VASEAAAKG
5]        pos= 229 score=0.018929 unnormalised=1.6735600000        VVLLGPGPG
6]        pos= 230 score=0.009632 unnormalised=0.8515600000        VLLGPGPGT
7]        pos= 246 score=0.033135 unnormalised=2.9295600000        FGPGPGACM
8]        pos= 259 score=0.010084 unnormalised=0.8915600000        LEKGPGPGL
9]        pos= 300 score=0.042409 unnormalised=3.7495600000        VTTQGPEAA
10]        pos=  -1 score=0.748392 unnormalised=66.1680000000        non-binding event
```